Reinforcement Learning IV
Optimal Value Functions

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What will be covered

- Optimal Value Functions
- Bellman Optimality Equations
- Optimal Polices

Recall: Value Functions

- $V^{\pi}(s)$: The expected return when starting in state s and following the policy π thereafter.
- $Q^{\pi}(s, a)$: The expected return when starting in state s, taking action a, and following the policy π thereafter.
- How can we evaluate policies to find the optimal policy π^{*}.

Recall: Policy Evaluation

$$V^{\pi}(s) = \sum_{a} \pi(a \mid s)[R(s,a) + \gamma V^{\pi}(s')]$$

What return do I get if I follow this policy?

- Computing the value of a fixed policy
- What if this policy isn't good (optimal)?

From Policy Evaluation to Optimal Control

Policy Evaluation	Optimal Control
Calculate the expected return if you follow π	Always choose the action with highest expected return
$V^{\pi}(s) = \sum_{a} \pi(a \mid s)[R + \gamma V^{\pi}(s')]$	$V^{*}(s) = \max_{a}[R + \gamma V^{*}(s')]$
$V^{\pi}(3,3) = 0.8(0.999) + 0.2(-0.001) + 0\,(-0.001) = 0.799$	$V^{*}(3,3) = \max\{0.999, -0{,}001, -0{,}001\} = 0.999$

Bellman Optimality Equations

- $V^*(s) = \max_a [R(s,a) + \gamma V^*(s')]$
  - Decomposes the value into the immediate reward and discounted future value

  What is the best return achievable from this state if I act optimally?

- $Q^*(s,a) = R(s,a) + \gamma \max_{a'} Q^*(s',a')$
  - The value of taking action a in state s, and then behaving optimally afterward.

  What is the best return achievable from this state if I first take action a and then act optimally therafter?

Optimal policy π^*

- How do we compare policies?
  - $\pi_1 \geq \pi_2 \; if \; V^{\pi_1}(s) \geq V^{\pi_2}(s) \; \forall s$
  - π_1 dominates π_2 if it yields higher expected return from every possible starting state
- $V^*(s) = \max_{\pi} V^{\pi}(s)$
  - value of s under the best policy among all possible policies.
- Are optimal policies unique?
- The optimal value function is unique but the optimal policy may not be!
  - Multiple actions can achieve the same maximum

V^* and Q^*

- $V^*(s)$: **Maximum** expected return achievable from state s when following an optimal policy π^*
- $Q^*(s, a)$: **Maximum** expected return starting from state s, taking action a, and **then** following an optimal policy π^*.

Finding an Optimal policy π^* from V^* and Q^*

- Once we have $V^*$and Q^* , we can obtain an optimal policy
- If we have Q^*
  - $\pi^*(s) = argmax\ Q^*(s, a)$
  - Choose the action with the highest optimal action-value.
- If we only have V^*
  - Perform a one-step lookahead using the model
  - $\pi^*(s) = \underset{a}{argmax}\, \mathbb{E}[R_{t+1} + \gamma V^*(s')]$

Argmax vs max

- $V^*(s) = \max_a Q^*(s,a) \rightarrow$ best value
  - What is the best possible return?

- $\pi^*(s) = argmax\ Q^*(s,a) \rightarrow$ best action
  - Which action gives the best return?

Solving MDPs and Bellman's equations

- Goal: Find or approximate the optimal policy π^* that maximises expected return.
- Use Bellman's optimality equations:
  - Solve for $V^*(s)$ or $Q^*(s, a)$ that satisfy the equations.
  - Extract π^* by choosing the best action in each state.
- Challenges:
  - Large or continuous state/action spaces make exact solutions impossible.
  - Iterative methods repeatedly update value estimates (V^* or Q^*) to gradually approach the optimal policy

Proof: Bellman Optimality Equation for V^*

Derivation of $V^*(s) = \max_a Q^*(s,a)$

- By definition: $V^*(s) = \max_\pi V^\pi(s) = \max_\pi \mathbb{E}_\pi[G_t \mid S_t = s]$
- $V^*(s) = \max_\pi \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s]$ --(recursive definition of G_t)
- $V^*(s) = \max_a \mathbb{E}\,[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a]$ --(optimise separately over the first action)
- $V^*(s) = \max_a [\mathbb{E}[\mathrm{R}_{t+1} | S_t = s, A_t = a] + \gamma \mathbb{E}[\mathrm{G}_{t+1} | S_t = s, A_t = a]]$ --(split expectation)

- $V^*(s) = \max_a \mathbb{E}[\mathrm{R}_{t+1} + \gamma V^*(S_{t+1}) \mid S_t = s, A_t = a]$ --(act optimally after the first action so $\mathbb{E}[G_{t+1} \mid S_{t+1} = s'] = V^*(s')$)

- $V^*(s) = \max_a \sum_{s',r} p\,(s', r|s, a)\,[r + \gamma V^*(s')]$ --(expand expectation using transition probabilities)
- $V^*(s) = \max_a Q^*(s,a)$ --(definition of Q^*)

Resources

- https://www.cs.cmu.edu/~mgormley/courses/10601-s17/slides/lecture26-ri.pdf
- https://spinningup.openai.com/en/latest/spinningup/rl_intro.html